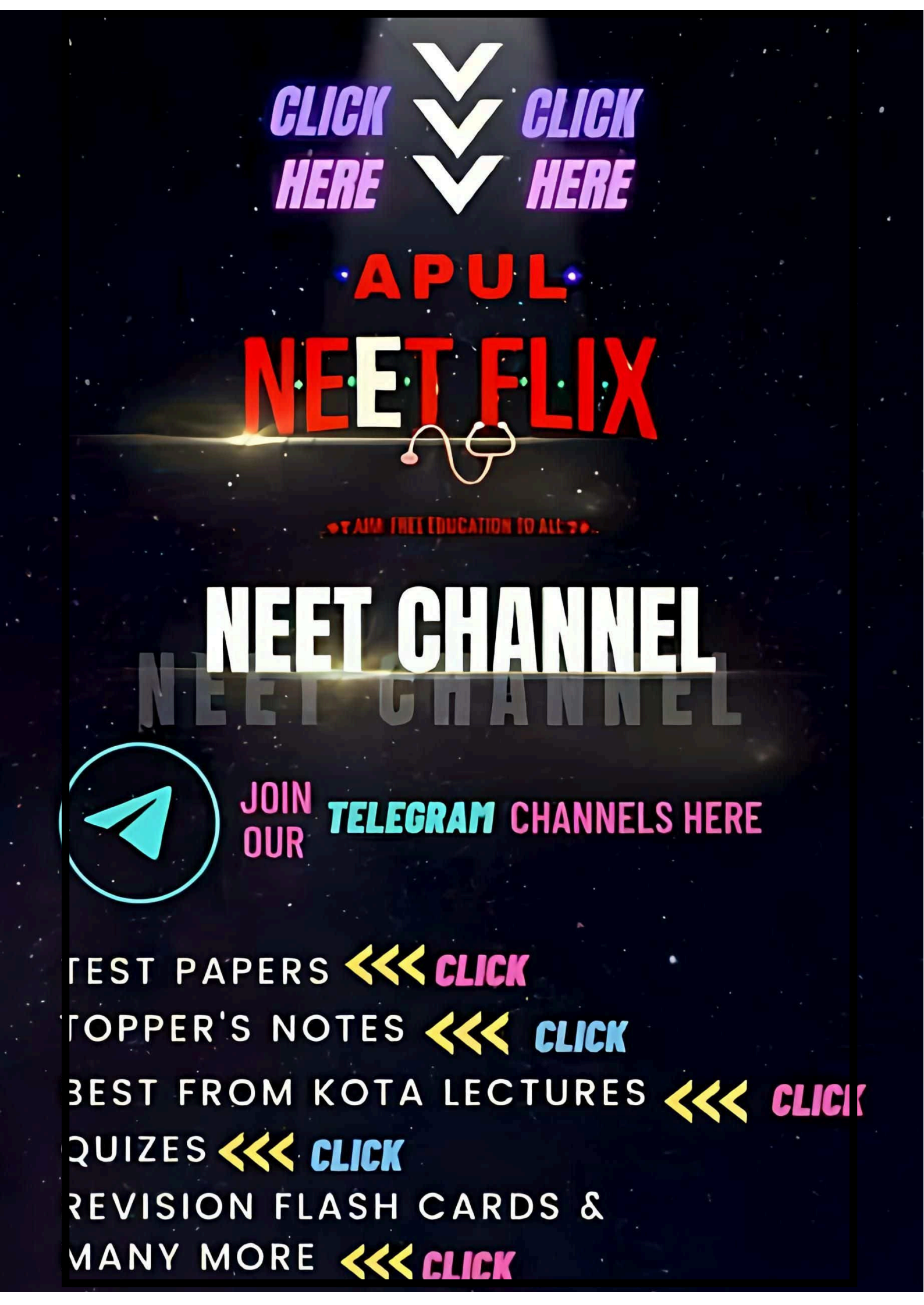

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PHYSICS

1)

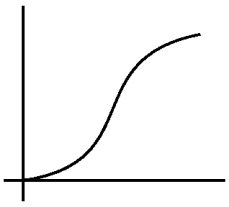
Maximum value of $\sqrt{7}\sin\theta + \sqrt{2}\cos\theta$ will be :-

(1) $\sqrt{5}$
(2) $\sqrt{7} + \sqrt{2}$
(3) 3
(4) None

2)

The value of $(\sin 15° + \cos 15°)^2$ is :-

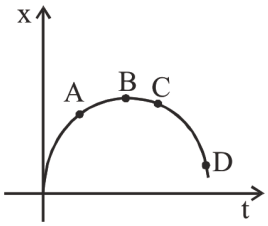
(1) 1
(2) $\frac{1}{2}$
(3) $\frac{3}{2}$
(4) $\frac{5}{2}$

3) The slope of the curve shown in figure :- 

(1) continuously increases
(2) continuously decreases
(3) first increases then decreases
(4) first decreases then increases

4)

Position-time graph of a particle is shown :-



Find point at which speed is maximum ?

(1) A

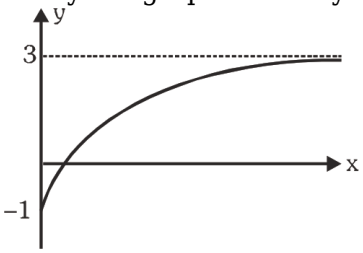
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(2) B

(3) C

(4) D

5) In the given figure a graph is shown a variable y varying exponentially with another variable x. Study the graph carefully. Which of the following equations best suits the above graph?



(1) $y = 3 - e^{-x}$

(2) $y = 1 - 4e^{-x}$

(3) $y = 1 - 3e^{-x}$

(4) $y = 3 - 4e^{-x}$

6) $y = \sqrt{6x+5}$, then $\frac{dy}{dx} =$

(1) $3\sqrt{6x+5}$

(2) $\frac{6}{\sqrt{6x+5}}$

(3) $\frac{3x}{\sqrt{6x+5}}$

(4) $\frac{3}{\sqrt{6x+5}}$

7) A particle moving along x-axis has, at any instant, its x-coordinate to be $x = 4 + 3t + 5t^2$. Then the value of $\frac{dx}{dt}$ at t = 1 is :-

(1) 12

(2) 13

(3) 7

(4) 8

8) Find $\frac{dy}{dx}$ if $y = \sin(4x-3)$

(1) $-\cos(4x-3)$

(2) $\frac{\cos(4x-3)}{4}$

(3) $\frac{\sin(4x-3)}{4}$

(4) $4\cos(4x-3)$

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9) If y = sin 2x – cos 2x then the maximum value of rate of change of y with respect to x is:-

(1) $\sqrt{6}$
(2) $\sqrt{8}$
(3) $\sqrt{20}$
(4) $\sqrt{24}$

10) A particle moves along the curve $12y = x^3$. Which coordinate changes at faster rate at x = 10 :-

(1) X - coordinate
(2) Y - coordinate
(3) Both X and Y - coordinate
(4) Data insufficient

11) A particle is moving along x-axis so that its x-coordinate varies with time 't' $x = 2–5t + 6t^2$ then initial velocity and acceleration of particle.

(1) –5 m/s; 12 m/s^2
(2) 12 m/s; –5m/s^2
(3) 6 m/s; –5 m/s^2
(4) 2 m/s; 6 m/s^2

12) Value of 4 + 8 + 12 + ... + 64 is :-

(1) 512
(2) 544
(3) 642
(4) 724

13) $\int \cos^2\theta \, d\theta$ is equal to :-

(1) $\frac{\theta}{2} + \frac{\sin 2\theta}{4} + c$
(2) $\frac{\sin 2\theta}{4} + c$
(3) $\frac{\theta}{4} - \frac{\cos 2\theta}{4} + c$
(4) $\frac{\theta}{2} - \frac{\sin 2\theta}{4} + c$

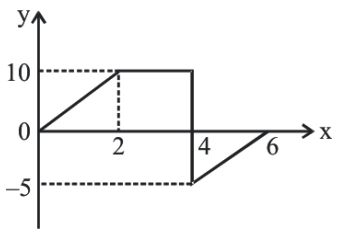
14) Value of $\int_0^{\frac{\pi}{2\omega}} 5\sin\omega t \; dt$ is :-

(1) 10 ω
(2) 5/ω

(3) $\frac{10\pi}{\omega}$

(4) $\frac{-5}{\omega}$

15) Value of $\int_0^6 ydx$ is :-



(1) 25

(2) 35

(3) 15

(4) 30

16) Determine the average value of y = 5x + 6 in the intervals $0 \le x \le 2$

(1) 11

(2) 13

(3) 15

(4) 17

17)

The initial velocity of a particle is u(at t = 0) and the acceleration is given by $a = 3t^2$. Which of the following is correct :-

(1) $v = u + \frac{1}{2}t^3$

(2) $v = u + t^3$

(3) $v = u + \frac{3t^3}{2}$

(4) None

18) A particle located at x = 0 at time t = 0, starts moving along the positive x-direction with a velocity 'v' that varies as $v = \alpha\sqrt{x}$. the velocity of the particle varies with time as :-

(1) t^2

(2) t

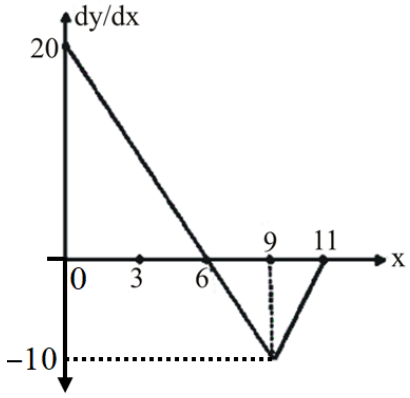
(3) $t^{1/2}$

(4) t^3

19) The following curve represent graph between rate of change of a variable y w.r.t x and x. The

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change in the value of y when x changes from 0 to 11 is :

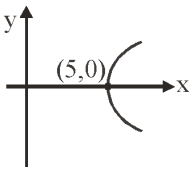
(1) 60
(2) 25
(3) 35
(4) 85

20) A body falls from a height of 3 m. If after every collision with ground it's maximum height becomes $\frac{1}{3}$ times, then find total distance travelled by body before coming to rest :-

(1) 4 m
(2) 6 m
(3) 8 m
(4) 9 m

21)

The equation of given graph is :-



(1) $y = -3x^2 - 5$
(2) $y = 3x^2 - 5$
(3) $x = -3y^2 + 5$
(4) $x = 3y^2 + 5$

22) The position vector of a particle is $\vec{r} = (a\cos\omega t)\hat{i} + (a\sin\omega t)\hat{j}$ The velocity of the particle is :-

(1) Parallel to position vector
(2) Perpendicular to position vector
(3) Directed towards the origin
(4) Directed away from the origin

23) A particle is moving 37° North of east with a velocity of 5 m/sec. In 10 sec the velocity change to 5 m/sec, 37° east of south. The magnitude of average acceleration in time is :-

(1) zero

(2) $\frac{1}{\sqrt{2}}m/sec^2$

(3) $5\sqrt{2}m/sec^2$

(4) $\frac{5}{\sqrt{2}}m/sec^2$

24) Which of the following sets of displacements might be capable of bringing a car to its starting point?

(1) 5, 9, 9 and 20 km

(2) 5, 10, 30 and 50 km

(3) 40, 40, 90 and 8 km

(4) 10, 20, 40 and 90 km

25) Two vectors $\vec{P} = 2\hat{i} + b\hat{j} + 2\hat{k}$ and $\vec{Q} = \hat{i} + \hat{j} + \hat{k}$ will be perpendicular if -

(1) b = 0

(2) b = 1

(3) b = 2

(4) b = – 4

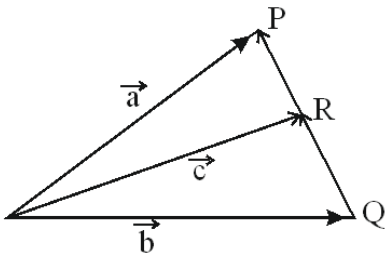
26) The adjacent sides of a parallelogram are represented by coinitial vectors $2\hat{i} + 3\hat{j}$ and $\hat{i} + 4\hat{j}$. The area of the parallelogram is : -

(1) 5 units along z-axis

(2) 5 units in x-z plane

(3) 3 units in x-z plane

(4) 3 units in y-x plane

27) 

The figure shows three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ where R is the mid-point of PQ. Then which of the following relation is correct ?

(1) $\vec{a} + \vec{b} = 2\vec{c}$

(2) $\vec{a} + \vec{b} = \vec{c}$

(3) $\vec{a} - \vec{b} = 2\vec{c}$

(4) $\vec{a} - \vec{b} = \vec{c}$

28) If $\vec{P} = \hat{i} + \hat{j} + 15\hat{k}$, $\vec{Q} = \hat{i} - 4\hat{j} + 3\hat{k}$ then unit vector along vector difference is :-

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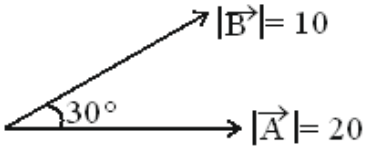
(1) $\hat{i}+\hat{j}+\hat{k}$

(2) $\frac{2\hat{i}-3\hat{j}+18\hat{k}}{\sqrt{324}}$

(3) $\frac{5\hat{j}+12\hat{k}}{13}$

(4) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

29) Which of the following is correct for given $\vec{A}$ and $\vec{B}$:



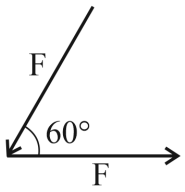
(1) $\vec{A}=\vec{B}$

(2) $|\vec{A}|=|\vec{B}|$

(3) $|\hat{A}|=|\hat{B}|$

(4) $\hat{A}=\hat{B}$

30) Two forces, each equal to F, act as shown in Fig. Their resultant is :-



(1) F/2

(2) F

(3) $\sqrt{3}F$

(4) $\sqrt{5}F$

31) Two balls are moving on the same smooth horizontal plane. Components of velocity of the first ball along x and y directions are $10\sqrt{3}$ and 30 m/s respectively. The components of velocity of the second ball along x and y directions are 40 and 30 m/s each. The angle between their directions of motion will be :-

(1) 7°

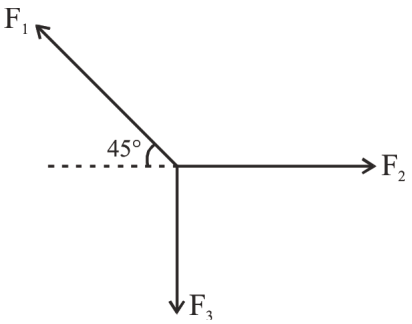
(2) 23°

(3) 27°

(4) 16°

32) Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.

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Column-I (Combination)		Column-II (Approximate Direction)	
i	$\vec{F}_1+\vec{F}_2+\vec{F}_3$	p	
ii	$\vec{F}_1-\vec{F}_2+\vec{F}_3$	q	
iii	$\vec{F}_1-\vec{F}_2-\vec{F}_3$	r	
iv	$\vec{F}_2-\vec{F}_1-\vec{F}_3$	s	

Now match the given columns and select the correct option from the codes given below.

	i	ii	iii	iv
(1)	p	q	r	s
(2)	s	r	p	q
(3)	q	r	p	s
(4)	r	s	p	q

(1) 1

(2) 2

(3) 3

(4) 4

33) If $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and if $\left|\hat{a}+\hat{b}\right|=\sqrt{3}$, then the value of $\left(2\hat{a}-5\hat{b}\right)\cdot\left(3\hat{a}+\hat{b}\right)$ is :-

(1) $\frac{41}{2}$

(2) $\frac{11}{2}$

(3) $-\frac{11}{2}$

(4) $-\frac{41}{2}$

34) Find the projection of $\vec{a}=2\hat{i}-\hat{j}+\hat{k}$ on $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$

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(1) $\frac{5}{\sqrt{6}}$

(2) $\frac{7}{10}$

(3) $\frac{6}{\sqrt{5}}$

(4) $\frac{5}{\sqrt{3}}$

35) How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant

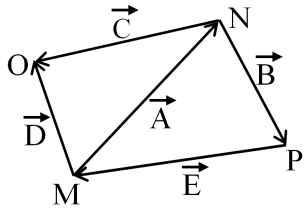
(1) 2

(2) 3

(3) 4

(4) 5

36) From figure, the correct relation is :-



(1) $\vec{A}+\vec{B}+\vec{E}=\vec{0}$

(2) $\vec{C}-\vec{D}=-\vec{A}$

(3) $\vec{B}+\vec{E}-\vec{C}=-\vec{D}$

(4) All of the above

37) Two vectors $\vec{A}$ and $\vec{B}$ lie in one plane, another vector $\vec{C}$ lie outside this plane, then $\vec{A}+\vec{B}+\vec{C}$:-

(1) Can be zero

(2) Can not be zero

(3) Lies in plane containing $\vec{A}+\vec{B}$

(4) Lies in plane containing $\vec{A}+\vec{C}$

38) Three vectors of equal magnitude F act at a point such that angle between every two vectors is 120°, their resultant is :

(1) F

(2) 2 F

(3) 3 F

(4) Zero

39) The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If $\vec{Q}$ is doubled then the new resultant vector is perpendicular to $\vec{P}$. Then magnitude of $\vec{R}$ is :–

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(1) $\frac{P^2 - Q^2}{2PQ}$

(2) Q

(3) $\frac{P}{Q}$

(4) $\frac{P + Q}{P - Q}$

40) $\vec{A} + \vec{B} = 2\hat{i}$ and $\vec{A} - \vec{B} = 4\hat{j}$ then what will be the angle between $\vec{A}$ and $\vec{B}$.

(1) $\cos^{-1}\left(\frac{-3}{5}\right)$

(2) $\cos^{-1}\left(\frac{3}{5}\right)$

(3) $\cos^{-1}\left(\frac{-4}{5}\right)$

(4) $\cos^{-1}\left(\frac{4}{5}\right)$

41) ABCDE is regular pentagon, O is its centre. If $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} = n(\overrightarrow{AO})$ then the value of 'n' is :-

(1) 3

(2) 4

(3) 5

(4) 6

42) If sum and difference of two vectors are perpendicular to each their then :-

(1) Vectors must be perpendicular

(2) Vectors must be equal

(3) Magnitude of vectors must be equal

(4) Vectors must be parallel

43) Velocity of a Projectile at time 't' is given by $\vec{V} = u\ \cos\theta\,\hat{i} + (u\sin\theta - gt)\hat{j}$. It initial velocity of projectile is $\vec{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$. then find time 't' when it will move perpendicular to in initial velocity. :-

(1) $\frac{u\sin\theta}{g}$

(2) $\frac{2u\sin\theta}{g}$

(3) $\frac{u}{g\sin\theta}$

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(4) $\frac{2u}{g\sin\theta}$

44) If $\vec{a},\vec{b},\vec{c}$ are three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, then $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ is equal to :

(1) – 1
(2) 3
(3) 0
(4) – 3/2

45) $\vec{F}_1+\vec{F}_2+\vec{F}_3=0$ and $\vec{F}_3$ is removed then acceleration of particle of mass m on which these are acting :-

(1) $\frac{\vec{F}_3}{m}$
(2) $\frac{-\vec{F}_3}{m}$
(3) $\frac{\vec{F}_2}{m}$
(4) $\frac{-\vec{F}_2}{m}$

CHEMISTRY

1) The rate equation for the reaction 2A + B → C is found to be : rate = k[A][B]. The correct statement in relation to this reaction is that :

(1) Rate of formation of C is twice the rate of disappearance of A.
(2) Half life is constant
(3) Unit of k must be s^{-1}
(4) Value of k is independent of the initial concentration of A and B

2) **Assertion :** On increasing temperature, rate of reaction increases.
Reason : On increasing temperature no. of activated molecules increases.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

3) Rate constant for a chemical reaction takes place at 500 K is expressed as $k = A.\ e^{-1000}$. The activation energy of the reaction is :

(1) 100 cal/mol
(2) 1000 kcal/mol

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(3) 10^4 kcal/mol

(4) 10^6 kcal/mol

4) In the Arrhenius plot of $\ln k$ vs $\frac{1}{T}$ linear plot is obtained with a slope of -2×10^4 K. Then energy of activation of the reaction (in kJ $mole^{-1}$) is (R = 8.3 JK^{-1} mol^{-1})

(1) 83

(2) 166

(3) 249

(4) 332

5) **Assertion :-** Order of reaction can never be fractional for an elementary reaction.
Reason :- Elementary reaction take place by multi step mechanism.

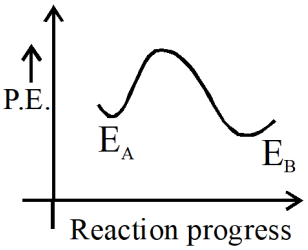
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

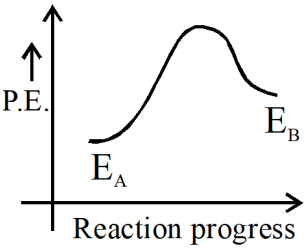
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

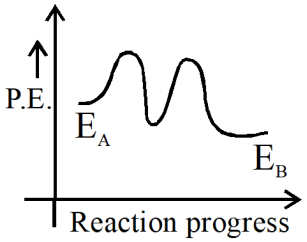
(3) If Assertion is True but the Reason is False.

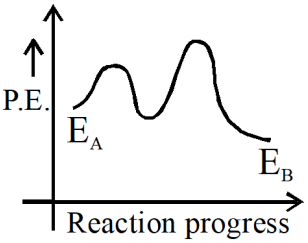
(4) If both Assertion & Reason are False.

6) For the reaction A → B, E_a = 10 kJ/mol and ΔH = 5kJ/mol. The correct potential energy diagram for the reaction can be :-

(1)


(2)


(3)


(4)


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7) Match the columns

	Column-I		Column-II
A	Number of collisions per second per unit volume of the reaction mixture.	p	Effective collisions.
B	Fraction of molecules with energies equal to or greater than E_a	q	Collision frequency
C	Arrhenius coefficient	r	$e^{-Ea/RT}$
D	Collision in which molecules collide with sufficient K.E. and proper orientation.	s	A

(1) A – (q), B – (r), C – (s), D – (p)

(2) A – (r), B – (q), C – (s), D – (p)

(3) A – (q), B – (s), C – (r), D – (p)

(4) A – (q), B – (r), C – (p), D – (s)

8) Match the columns

	Column-I		Column-II
A	The decomposition of gaseous ammonia on a hot platinum surface	p	Zero order reaction
B	The thermal decomposition of HI on gold surface	q	Pseudo first order reaction
C	All natural and artificial radioactive decay of unstable nuclei	r	Zero order reaction at high pressure
D	Inversion of cane sugar	s	First order reaction.

(1) A – (r), B – (p), C – (s), D – (q)

(2) A – (r), B – (s), C – (q), D – (p)

(3) A – (q), B – (s), C – (p), D – (r)

(4) A – (q), B – (p), C – (s), D – (r)

9) K for a zero order reaction is 2×10^{-2} mol L^{-1} sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M, the initial concentration must have been.

(1) 0.5 M

(2) 1.25 M

(3) 12.5 M

(4) 1.0 M

10) If a first order reaction is completed to the extent of 75% and 50% in time intervals, t_1 and t_2, what is the ratio, $t_1 : t_2$?

(1) ln 2

(2) $\frac{\ln(3/4)}{\ln 2}$

(3) 2

(4) 1/2

11) In a first order reaction the reacting substance has half-life period of ten minutes. What fraction of the substance will be left after an hour the reaction has occurred?

(1) 1/6 of initial concentration

(2) 1/64 of initial concentration

(3) 1/12 of initial concentration

(4) 1/32 of initial concentration

12) Which is not true for a second order reaction ?

(1) It can have rate constant 1×10^{-2} L mol^{-1} s^{-1}

(2) Its half-life is inversely proportional to its initial concentration of reactant

(3) Time to complete 75% reaction is twice of half-life

(4) $t_{50\%} = \frac{1}{[A]_0 k}$

13) For the reaction H_2 (g) + Br_2 (g) → 2HBr (g), the experiment data suggested that $r = k[H_2][Br_2]^{1/2}$ The molecularity and order of the reaction are respectively

(1) 2, 3/2

(2) 3/2, 3/2

(3) Not defined, 3/2

(4) 1, 1/2

14)

A zero order reaction A → B completed 30% in 60 minute when initial concentration of A is taken as 0.1 $\frac{Mol}{L}$. Then what will be the concentration of B at 250 minute.

(1) 0.08 M

(2) 0.1 M

(3) 0.125 M

(4) 0.0125 M

15) A hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below :

$A_2 \rightleftharpoons A + A$ (fast)
$A + B_2 \rightarrow AB + B$ (slow)
$A + B \rightarrow AB$ (fast)
The order of the overall reaction is :

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(1) 2

(2) 1

(3) 1.5

(4) 0

16)

In the following reaction xA → yB

$$\log_{10}\left[\frac{-d[A]}{dt}\right] = \log_{10}\left[\frac{d[B]}{dt}\right] + 0.3010$$

A and B respectively can be -

(1) n-butane and isobutane

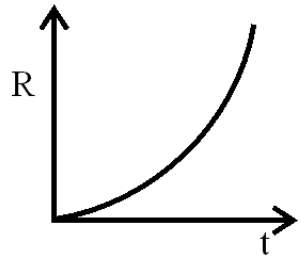
(2) C_2H_2 and C_6H_6

(3) C_2H_4 and C_4H_8

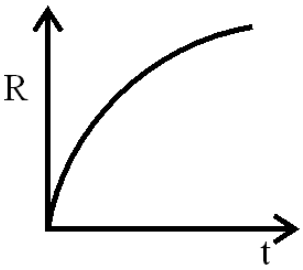
(4) N_2O_4 and NO_2

17) The reaction X(g) → Y(g) follows first order kinetics. The correct graph representing the rate of formation (R) of Y(g) with time (t) is.

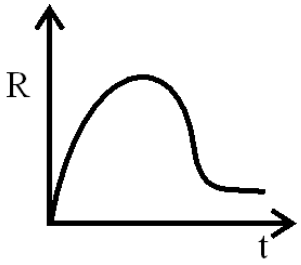
(1)



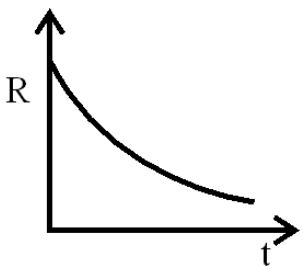
(2)



(3)



(4)



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18) The rate of reaction at 273 K is R_0. The rate of reaction at 313 K will be (Assuming temperature coefficient equal to 2) –

(1) 16 R_0

(2) 64 R_0

(3) $\frac{R_0}{32}$

(4) $\frac{R_0}{16}$

19) For reaction A → B , the rate constant $k_1 = A_1\left(e^{-E_{a_1}/RT}\right)$ and for the reaction X → Y, the rate constant $k_2 = A_2\left(e^{-E_{a_2}/RT}\right)$. If $A_1 = 10^9$, $A_2 = 10^{10}$ and E_{a1} = 1200 cal/mol and E_{a2}=1800 cal/mol, then the temperature at which $k_1 = k_2$ is :
(Given ; R = 2 cal/K-mol)

(1) 300 K

(2) 300 × 2.303 K

(3) $\frac{300}{2.303}$K

(4) None of these

20) For the reaction, 2A → B + 3C;
if $-\frac{d[A]}{dt} = k_1[A]^2$; $\frac{d[B]}{dt} = k_2[A]^2$; $\frac{d[C]}{dt} = k_3[A]^2$ the correct relation between k_1, k_2 and k_3 is :-

(1) $k_1 = k_2 = k_3$

(2) $2k_1 = k_2 = 3k_3$

(3) $4k_1 = k_2 = 3k_3$

(4) $\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

21) For the reaction, $2N_2O_{5(g)} \rightarrow 4\ NO_{2(g)} + O_{2(g)}$, the rate of formation of O_2 is $0.032 ghr^{-1}$, then rate of conversion of N_2O_5 in ghr^{-1} is :-

(1) 0.216

(2) 0.098

(3) 0.627

(4) 0.313

22) The mechanism of the reaction $2NO_2 + F_2 \rightarrow 2NO_2F$ is :-

(i) $NO_2 \xrightarrow{\text{Slow}} NO + O$

(ii) $F_2 + O + NO \xrightarrow{\text{fast}} NO_2F + F$

(iii) $F + NO_2 \xrightarrow{\text{fast}} NO_2F$.

Select the correct one

(1) the reaction is of 3rd order

(2) the molecularity of the reaction is sum of molecularities of all steps

(3) reaction is zero order w.r.t. F_2

(4) half life of reaction depends upon initial conc. of NO_2.

23) In the given first order reaction
$NH_4NO_{2(aq.)} \rightarrow N_{2(g)} + 2H_2O_{(\ell)}$ the volume of N_2 after 20 min and after a long time is 40 ml and 70 ml respectively. The value of rate constant is:-

(1) (1/20) ℓn (7/3) min^{-1}

(2) (2.303/1200) log (7/4) sec^{-1}

(3) (1/20) log (7/3) min^{-1}

(4) (2.303/20) log (11/7) min^{-1}

24) Which of the following statement is incorrect ?

(1) $t_{1/2}$ (half life) increases with increase in temperature.

(2) Zero order reaction is a complex reaction.

(3) Rate constant increases with increase in temperature.

(4) Both (1) & (3)

25) For a first order gaseous reaction, $A(g) \rightarrow B(g) + C(g)$
If P_i is the initial pressure of A and P_t is total pressure at time t. Then formula of rate constant is -

(1) $k = \frac{2.303}{t} \log \frac{P_i}{P_i - P_t}$

(2) $k = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_t}$

(3) $k = \frac{1}{t} \log \frac{P_i}{2P_i - P_t}$

(4) $k = \frac{2.303}{t} \log \frac{2P_i}{2P_i - P_t}$

26) For first order reaction, the slope of graph between $\log \frac{[R]_0}{[R]}$ vs t is -

(1) $\frac{2.303}{K}$

(2) $\frac{K}{2.303}$

(3) 2.303 K

(4) $\frac{1}{K}$

27) The following results have been obtained during the kinetics studies of the reaction $2A + B \rightarrow C + D$

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Experiment	[A]	[B]	Initial rate of formation of D (mol L^{-1} min^{-1})
I	0.1	0.1	6×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

The rate law for the reaction will be :-

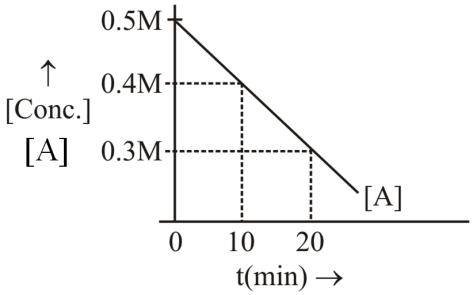
(1) $r = k[A]\,[B]^2$

(2) $r = k[A]^2\,[B]$

(3) $r = k[A]\,[B]$

(4) $r = k[A]^{1/2}\,[B]$

28) 2A → Products



Calculate average rate of reaction during 0 to 10 min. and order of reaction

(1) 0.010 M min^{-1}, 1st order

(2) 0.005 M min^{-1}, 0 order

(3) 0.005 M min^{-1}, 1st order

(4) 0.010 M min^{-1}, 0 order

29) Which of the following statement is false for a zero order reaction.

(1) Unit of rate of reaction & rate constant is same

(2) $t_{75\%} = 3/2\ t_{50\%}$

(3) Slope of graph between $[A]_t$ vs t is –K

(4) On increasing the value of K, half life of zero order reaction will also increase

30) 90 % of a first order reaction was completed in 10 hours. When will 99.9 % of the reaction complete?

(1) 20 hr

(2) 30 hr

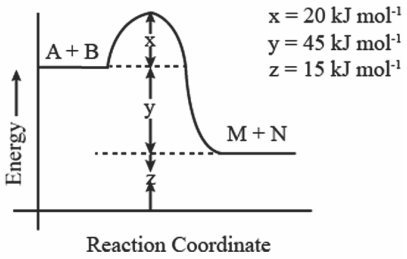
(3) 40 hr

(4) 35 hr

31) According to the following figure, the magnitude of the enthalpy change of the reaction A +

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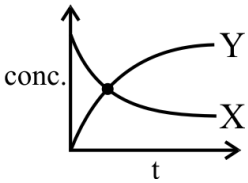
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B→M + N is equal to.....(kJ/mol)

(1) 20
(2) 45
(3) 15
(4) 65

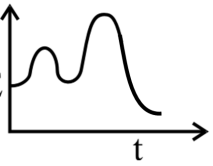
32) 2X → Y

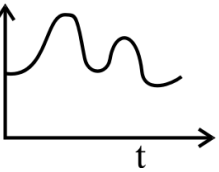


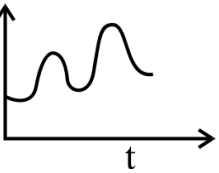
life period of reaction at point of intersection is –

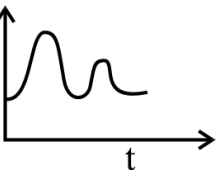
(1) $t_{1/2}$
(2) $t_{1/3}$
(3) $t_{2/3}$
(4) $t_{3/4}$

33) Which of the following will have second step as slow step in exothermic reaction?

(1)


(2)


(3)


(4)


34) A and B are two substances undergoing first order decomposition with half lives 0.2 sec and 0.3 sec. respectively. If the initial concentration of A and B are in the ratio 3 : 2 then the ratio of concentrations of A and B after three half lives of A is :-

(1) 4 : 3

(2) 3 : 2

(3) 3 : 4

(4) 2 : 3

35) Match the columns

	Column-I		Column-II
(A)	Mathematical expression for rate of reaction	(p)	rate constant
(B)	Rate of reaction for zero order reaction is equal to	(q)	rate law
(C)	Units of rate constant for zero order reaction is same as that of	(r)	order of slowest step
(D)	Order of a complex reaction is determined by	(s)	rate of reaction

(1) A-(q),B-(p),C-(s),D-(r)

(2) A-(r),B-(p),C-(s),D-(q)

(3) A-(q),B-(s),C-(p),D-(r)

(4) A-(p),B-(q),C-(s),D-(r)

36) For the reaction 2A + B → product, if rate of disappearance of A is $2 \times 10^{-4} \frac{\text{mol}}{\text{L} - \text{sec}}$ then rate of disappearance of B will be

(1) $\frac{1}{2} \times 10^{-4} \frac{\text{mol}}{\text{L} - \text{sec}}$

(2) $3 \times 10^{-4} \frac{\text{mol}}{\text{L} - \text{sec}}$

(3) $4 \times 10^{-4} \frac{\text{mol}}{\text{L} - \text{sec}}$

(4) $10^{-4} \frac{\text{mol}}{\text{L} - \text{sec}}$

37) The energy of activation for forward and backward change for an endothermic reaction : x → y are E_f and E_b respectively. Which of the following is correct?

(1) $E_b < E_f$

(2) $E_b > E_f$

(3) $E_b = E_f$

(4) No relation between them

38) **Statement – I :** Zero order reaction is a complex reaction.
Statement – II : A bimolecular elementary reaction is a second order reaction.
On the basis of two statements select the right option.

(1) Statement I is incorrect, statement II is correct

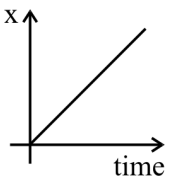
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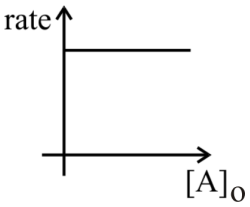
(2) Statement I is correct, statement II is incorrect

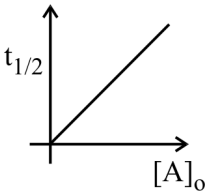
(3) Both statements are correct

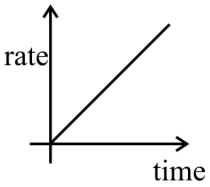
(4) Both statements are incorrect

39) Which graph is incorrect for a zero order reaction.

(1) 

(2) 

(3) 

(4) 

40) For a reaction A → 2B + C the half lives are 100 s and 50 s when the concentration of reactant A is 0.5 and 1.0 mol L^{-1} respectively. The order of the reaction is.....

(1) 0

(2) 2

(3) 1

(4) 3

41) **Assertion :** The reaction having low value of activation energy are generally fast.
Reason : Temperature coefficient of reaction having low activation energy is large.

(1) Both Assertion & Reason are True &the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are false.

42) Which of the following statements are applicable to a balanced chemical equation of an elementary reaction ?
(i) Order is same as molecularity
(ii) Order is less than the molecularity
(iii) Order is greater than the molecularity

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(iv) Molecularity can never be zero

(1) ii, iii
(2) i, iv
(3) i,ii
(4) ii,iv

43) The activation energy for the reaction,
$2HI(g) \rightarrow H_2(g) + I_2(g)$,
is 209.5 kJ at 581 K. Calculate the fraction of molecules of reactants having energy equal to or greater than activation energy.

(1) 1.5×10^{-19}
(2) 1.4×10^{-18}
(3) 1.4×10^{-17}
(4) 1.4×10^{-20}

44) The activation energies of two reactions are E_1 and E_2 ($E_1 > E_2$). If the temperature of the system is increased from T_1 to T_2, the rate constant of the reactions changes from k_1 to k'_1 in the first reaction and k_2 to k'_2 in the second reaction. Predict which of the following expression is correct ?

(1) $\frac{k'_1}{k_1} = \frac{k'_2}{k_2}$
(2) $\frac{k'_1}{k_1} > \frac{k'_2}{k_2}$
(3) $\frac{k'_1}{k_1} < \frac{k'_2}{k_2}$
(4) $\frac{k'_1}{k_1} = \frac{k'_2}{k_2} = 1$

45) For $A_{(g)} + B_{(g)} \rightarrow C_{(g)}$:
rate = $K[A]^{1/2}[B]^2$. If initial concentration of A and B are increased by a factor of 4 and 2 respectively, the initial rate is changed by the factor.

(1) 2
(2) 4
(3) 6
(4) 8

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**3**	**3**	**3**	**4**	**4**	**4**	**2**	**4**	**2**	**2**	**1**	**2**	**1**	**2**	**1**	**1**	**2**	**2**	**3**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**4**	**2**	**2**	**1**	**4**	**1**	**1**	**3**	**3**	**2**	**2**	**3**	**3**	**1**	**2**	**4**	**2**	**4**	**2**	**1**
Q.	41	42	43	44	45															
A.	**3**	**3**	**3**	**4**	**2**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**4**	**1**	**2**	**2**	**3**	**2**	**1**	**1**	**4**	**3**	**2**	**3**	**3**	**2**	**3**	**3**	**4**	**1**	**3**	**4**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**1**	**3**	**1**	**1**	**2**	**2**	**1**	**2**	**4**	**2**	**2**	**3**	**1**	**3**	**1**	**4**	**1**	**3**	**4**	**2**
Q.	86	87	88	89	90															
A.	**3**	**2**	**1**	**2**	**4**															

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PHYSICS

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1) $y = \sqrt{7}\sin\theta + \sqrt{2}\cos\theta$

$y_{max} = \sqrt{(\sqrt{7})^2 + (\sqrt{2})^2} = 3$

2)

Let α = sin 15º and b = cos 15º.

$(\sin 15º + F \cos 15º)^2 = \sin^2 15º + 2 \sin 15º \cos 15º + \cos^2 15º$

Using the identity $\sin^2 \theta + \cos^2 \theta = 1$ and $2 \sin \theta \cos \theta = \sin 2\theta$, we get:

$= 1 + \sin 30º$

Since $\sin 30º = \frac{1}{2}$, the expression becomes:

$1 + \frac{1}{2} = \frac{3}{2}$

3)

At first $\frac{dy}{dx}$ is increasing then decreasing.

4)

Magnitude of slope of x-t graph shows speed.

5) Shift the curve $(-4e^{-x})$ in positive y-direction by 3 units.

Only option 4 satisfy the (0, –1).

6) $y = \sqrt{6x + 5}$

$\frac{dy}{dx} = \frac{1(6)}{2\sqrt{6x + 5}} = \frac{3}{\sqrt{6x + 5}}$

7)

$x = 4 + 3t + 5t^2$

$\frac{dx}{dt} = 3 + 10t$

at t = 1

$\frac{dx}{dt} = 3 + 10 = 13$

8)

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$y = \sin(4x - 3)$

$\frac{dy}{dx} = \cos(4x - 3)(4)$

$= 4\cos(4x - 3)$

9)

$y = \sin 2x - \cos 2x$

$\frac{dy}{dx} = 2\cos 2x + 2\sin 2x$

$\left(\frac{dy}{dx}\right)_{max} = \sqrt{2^2 + 2^2} = \sqrt{8}$

10)

$12y = x^3$

$12\frac{dy}{dt} = 3x^2\frac{dx}{dt}$

$\frac{dy}{dt} = \left(\frac{x^2}{4}\right)\frac{dx}{dt}$

at $x = 10$

$\frac{dy}{dt} > \frac{dx}{dt}$

⇒ y-coordinate change at faster rate than π-coordinate

11) $v = \frac{dx}{dt} = -5 + 12t$

$a = \frac{dv}{dt} = 12$

12)

$y = 4 + 8 + 12 + ... + 64$

$64 = 4 + (n - 1) \times 4$

$n = 16$

$S = \frac{16}{2}(4 + 64) = 544$

13) $\int \cos^2\theta \, d\theta$

$\int \left(\frac{1 + \cos 2\theta}{2}\right) d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + c$

14) $\int_0^{\frac{\pi}{2\omega}} 5\sin\omega t \, dt = \frac{-5}{\omega}[\cos\omega t]_0^{\frac{\pi}{2\omega}}$

$\Rightarrow = \frac{-5}{\omega}\left[\cos\frac{\pi}{2} - \cos 0\right] = \frac{5}{\omega}$

15) $\int_0^6 y\,dx$ = Area under the curve

$$< y >= \frac{\int_{x_1}^{x_2} y dx}{\int_{x_1}^{x_2} dx} = \frac{\int_0^2 (5x+6)dx}{\int_0^2 dx}$$

$$= \frac{5\left[x^2/2\right]_0^2 + 6(x)_0^2}{(x)_0^2}$$

$$= \frac{\frac{5}{2}[4-0]\ \ 6(2-0)}{(2-0)}$$

$$= \frac{10+12}{2} = 11$$

16)

17)

$a = 3t^2$

$\frac{dv}{dt} = 3t^2$

$\int_u^v dv = \int_0^t 3t^2 dt$

$v - u = \frac{3t^3}{3}$

$\{v = u + t^3\}$

18)

Compare $v = \alpha\sqrt{x}$ with

$v = \sqrt{2ax}$

$\Rightarrow$ acceleration = $\frac{\alpha^2}{2}$ $\Rightarrow v = at$ $v \propto t$

19) $\text{Area} = \int \frac{dy}{dx} \cdot dx$

$= \int dy$

$= \Delta y$

So, $y = \text{Area} = \frac{1}{2} \times 20 \times 6 - \frac{1}{2} \times 10 \times 5$

$= 60 - 25$

$= 35$

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20) $S = h + \frac{2h}{3} + \frac{2h}{9} + ...$

$S = h + \frac{2h}{3}\left(1 + \frac{1}{3} + ...\right)$

$S = h + \frac{2h}{3}\left(\frac{1}{1-\frac{1}{3}}\right)$

$S = h + \frac{2h}{3} \times \frac{3}{2} = 2h = 6m$

21) $x = 3y^2 + 5$ (From options)

22) $\vec{r} = (a\cos\omega t)\hat{i} + (a\sin\omega t)\hat{j}$

$\vec{v} = \frac{d\vec{r}}{dt} = -a\omega\sin\omega t\hat{i} + a\omega\cos\omega t\hat{j}$

$\vec{r} \cdot \vec{v} = 0$

23) $\vec{a} = \frac{5\hat{j} - 5\hat{i}}{10}$

$\vec{a} = \frac{1}{2}(\hat{j} - \hat{i})$

$|a| = \frac{1}{2}\sqrt{2}$

$= \frac{1}{\sqrt{2}} m/sec^2\ (N - W)$

24)

Maximum value of addition of smallest three displacement should be greater than equal to the remaining fourth displacement.

25)

If $\vec{P} \perp \vec{Q}$ then $\vec{P}.\vec{Q} = 0$

$(2\hat{i} + b\hat{j} + 2\hat{k}).(\hat{i} + \hat{j} + \hat{k}) = 0$

$2 + b + 2 = 0$

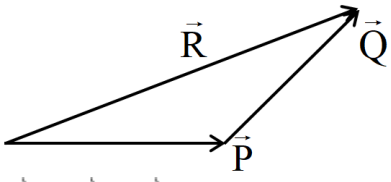
$b = -4$

26) $\vec{A} = (2\hat{i} + 3\hat{j}) \times (\hat{i} + 4\hat{j})$

$= (8\hat{k} - 3\hat{k}) = 5\hat{k}$

27) **Explanation:** In this question we have to choose the correct relation for the given figure by Triangle law of vector addition.

Concept: If two vectors drawn in the same order are representing two sides of a triangle, then the third side of the triangle drawn from the tail of one vector to the head of another vector represents the resultant of both the vectors.



$\vec{R} = \vec{R} + \vec{Q}$

If two vectors have the same direction and the same magnitude, then both the vectors are equal.

Calculations: As per the given figure

R is the Mid of $\overrightarrow{PQ}$

$\Rightarrow \overrightarrow{QR} = \overrightarrow{RP}$

Now by triangle law of vector addition

$\Rightarrow \vec{b} + \overrightarrow{QR} = \vec{c}$ – (1)

$\Rightarrow \vec{c} + \overrightarrow{RP} = \vec{a}$ - (2)

$\Rightarrow \overrightarrow{RP} = \vec{a} - \vec{c}$ - (3)

Now since $\overrightarrow{OP} = \overrightarrow{RP} \Rightarrow \overrightarrow{QR} = \vec{a} - \vec{c}$

28) $\vec{P} - \vec{Q} = 0\hat{i} + 5\hat{j} + 12k = \vec{R}$

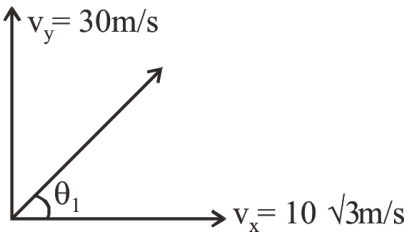
$\hat{R} = \frac{5\hat{j} + 12\hat{k}}{13}$

29) $\vec{A}$ and $\vec{B}$ have different magnitude and different directions so only $|\hat{A}| = |\hat{B}| = 1$ is correct.

30) $\theta = 120°$

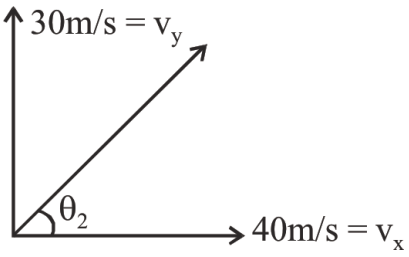
$R = 2a\cos\left(\frac{\theta}{2}\right)$

31)



$\tan\theta_1 = \frac{v_y}{v_x} = \frac{30}{10\sqrt{3}} = \sqrt{3}$

$\theta_1 = 60°$



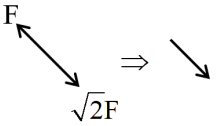
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$\tan\theta_2 = \frac{30}{40} \Rightarrow \theta_2 = \tan^{-1}\left(\frac{3}{4}\right),\ \theta_2 = 37^\circ$

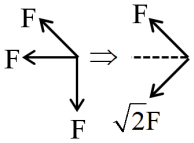
$\theta_1 - \theta_2 = 60^\circ - 37^\circ$

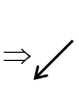
$\boxed{\theta_1 - \theta_2 = 23^\circ}$

32)

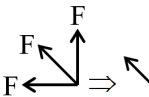
(i) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 \Rightarrow$ 

$\Rightarrow$ (q)

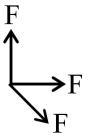
(ii) $\vec{F}_1 - \vec{F}_2 + \vec{F}_3 \Rightarrow$ 

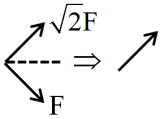
$\Rightarrow$ 

$\Rightarrow$ (r)

(iii) $\vec{F}_1 - \vec{F}_2 - \vec{F}_3 \Rightarrow$ 

$\Rightarrow$  (p)

(iv) $\vec{F}_2 - \vec{F}_1 - \vec{F}_3 \Rightarrow$ 

$\Rightarrow$ 

$\Rightarrow$ (s)

33) $|\hat{a} + \hat{b}| = \sqrt{3}$

$\theta \neq 0, 180^\circ$

$3 = 1 + 1 + 2\hat{a}\cdot\hat{b}$

$\hat{a}\cdot\hat{b} = \frac{1}{2}$

$(2\hat{a} - 5\hat{b})\cdot(3\hat{a} + \hat{b}) = 6 - 5 - 13\hat{a}\cdot\hat{b}$

$= 1 - \frac{13}{2} = -\frac{11}{2}$

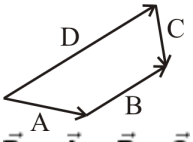
34) $\left|\vec{a}_{||\vec{b}}\right| = \frac{\vec{a}.\vec{b}}{|\vec{b}|} = \frac{(2\hat{i} - \hat{j} + \hat{k})(\hat{i} - 2\hat{j} + \hat{k})}{|\hat{i} - 2\hat{j} + \hat{k}|}$

$= \frac{2 + 2 + 1}{\sqrt{6}} = \frac{5}{\sqrt{6}}$

35)

If magnitudes are different, minimum three coplanar vectors are required to give zero resultant.

36) For a closed polygon

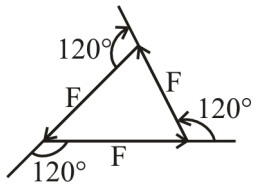


$\vec{D} = \vec{A} + \vec{B} + \vec{C}$

37)

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If vectors are non-coplanar then minimum four vectors are required to give zero resultant.

38) 

Zero

39) $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$\tan\alpha = \tan 90° = \frac{2(2Q)}{P + 2Q\cos\theta} = \infty$

$P + 2Q\cos\theta = 0$

$\cos\theta = \frac{-P}{2Q}$

From (1) & (2)

$R^2 = P^2 + Q^2 + 2PQ\left(\frac{-P}{2Q}\right)$

$R = Q$

$\vec{A} + \vec{B} = 2\hat{i}$

$\vec{A} - \vec{B} = 4\hat{j}$

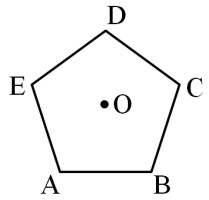
$\vec{A} = \hat{i} + 2\hat{j}$

40) $\vec{B} = \hat{i} - 2\hat{j}$

Now $\vec{A} \cdot \vec{B} = AB\cos\theta$

$-3 = \sqrt{5}\sqrt{5}\cos\theta$

$\cos\theta = \frac{-3}{5}$

41) 

$\vec{R} = \overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE}$

$\vec{R} = (\overrightarrow{AO} + \overrightarrow{OB}) + (\overrightarrow{AO} + \overrightarrow{OC}) + (\overrightarrow{AO} + \overrightarrow{OD}) + (\overrightarrow{AO} + \overrightarrow{OE})$

$\vec{R} = 5\overrightarrow{AO} + (\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} + \overrightarrow{OE})$

$\vec{R} = 5\overrightarrow{AO}$

42) $(\vec{A} + \vec{B}).(\vec{A} - \vec{B}) = 0$

$A^2 - B^2 = 0$

$A = B$

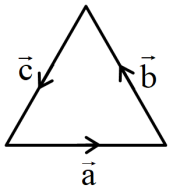
43) $\vec{u}.\vec{v} = 0$

$(u\cos\theta\hat{i} + u\sin\theta\hat{j}) \cdot (u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}) = 0$

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$u^2\cos^2\theta + u^2\sin^2\theta - u\sin\theta gt = 0$
$u^2 - u\sin\theta gt = 0$
$t = \frac{u}{g\sin\theta}$

44) $|\vec{a}| = |\vec{b}| = |\vec{c}| = 0$
and $\vec{a} + \vec{b} + \vec{c} = 0$



angle between any two vector will be 120°
Now $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = 3\cos\theta = -\frac{3}{2}$

45) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$
$\vec{F}_1 + \vec{F}_2 = -\vec{F}_3$ ...(1)
If $\vec{F}_3$ is removed $\vec{a} = \frac{\vec{F}_1 + \vec{F}_2}{m} = \frac{-\vec{F}_3}{m}$

CHEMISTRY

46)

$2A + B \rightarrow C$
$\frac{(ROD)A}{2} = (ROA)C$
$\Rightarrow r = k[A][B]$
order = 2
$t_{1/2} = \frac{1}{[A]_0 k}$
unit of k : $mol^{-1}\ L\ s^{-1}$

47) **Assertion:** On increasing temperature, the rate of reaction increases.
$k = Ae^{-Ea/RT}$
Increasing T increases k, leading to a faster reaction rate.
Reason: On increasing temperature, the number of activated molecules increases.
Boltzmann distribution shows that at **higher T,** more molecules have energy $\geq$ **activation energy (E_a),** increasing successful collisions.

48)

$\Rightarrow \frac{Ea}{RT} = 1000$
Ea = 1000 × 2 × 500 cal

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Ea = 10^6 cal = 10^3 kcal/mol

49)

Slope = $\frac{-E_a}{R}$

$\frac{-E_a}{R} = -2 \times 10^4$

$Ea = 2 \times 10^4 R = \frac{2 \times 10^4 \times 8.3}{1000} = 166$

50)

Elementary reactions completes in a single step.

51)

$\Delta H = (E_a)_f - (E_a)_b$

52)

Collision frequency → Number of collision per Second Per units Volume of reaction mixture
Fraction of molecule with energy equal to or greater than Ea → $e^{-Ea/RT}$
Arrhenius Coefficient → A

53)

Surface catalyzed reactions at high pressures often show zero order kinetics. Radioactive decay is one of the simplest first order processes known. In reactions in aqueous solution with excess solvent, pseudo first order is observed.

54)

$[A]_0 = kt + [A]_t$
$[A]_0 = 0.5 + 2 \times 10^{-2} \times 25$
$[A]_0 = 0.5 + 0.5 = 1M$

55)

For 75% completion ($\alpha_1 = 0.75$):
$t_1 = \frac{1}{k} \ln\left(\frac{1}{1-0.75}\right) = \frac{1}{k} \ln 4 = \frac{2 \ln 2}{k}$
For 50% completion ($\alpha_2 = 0.50$):
$t_2 = \frac{1}{k} \ln\left(\frac{1}{1-0.50}\right) = \frac{1}{k} \ln 2 = \frac{\ln 2}{k}$
Therefore, the ratio:

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$$t_1 : t_2 = \frac{2\ln 2/k}{\ln 2/k} = 2 : 1$$

56) $[A]_t = \frac{[A]_0}{2^n}$

No. of Half lives = $\frac{60}{10} = 6 = n$

$[A]_t = \frac{[A]_0}{2^6} = \frac{[A]_0}{64}$

57) $t_{1/2} = \frac{1}{[A]_0 k}$

58)

$r = k[H_2]\,[Br_2]^{1/2}$

order = $1 + \frac{1}{2} = \frac{3}{2}$

59)

Reaction completed in 200 minute

60)

$r = k\,[A]\,[B_2]$

$k_{eq} = \frac{[A]^2}{[A_2]}$

$[A] = k_{eq}\,[A_2]^{1/2}$

$r = k[A_2]^{1/2}\,[B_2]$

order = $1 + \frac{1}{2} = \frac{3}{2}$

61) $\log\left(\frac{-d[A]}{dt}\right) = \log\left(\frac{d[B]}{dt}\right) + \log 2$

$\log\left(\frac{-d[A]}{dt}\right) = \log\left(\frac{2d[B]}{dt}\right)$

$\frac{-d[A]}{dt} = \frac{2d[B]}{dt}$

$\Rightarrow \frac{1}{2}\left(\frac{-d[A]}{dt}\right) = \frac{d[B]}{dt}$

$\Rightarrow 2A \rightarrow B$

$\Rightarrow 2C_2H_4 \rightarrow C_4H_8$

62)

$[A]_t = [A]_0\, e^{-kt}$

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63) $R = R_0(2)^{\frac{313-273}{10}}$

$R = R_0(2)^4$

$R = 16\ R^0$

64)

$K_1 = K_2$

$A_1e^{-Ea_1/RT} = A_2e^{-Ea_2/RT}$

$\frac{e^{-Ea_1/RT}}{e^{-Ea_2/RT}} = A_2/A_1$

$e^{\frac{1}{RT}(Ea_2-Ea_1)} = 10^{10}/10^9 = 10$

$e^{\frac{1800-1200}{RT}} = 10$

$e^{600/RT} = 10$

$\frac{600}{RT} = \log_e 10$

$\frac{600}{2 \times T} = 2.303 \times 1$

$T = \frac{300}{2.303}$

65) $\frac{1}{2}k_1[A]^2 = k_2[A]^2 = \frac{1}{3}k_3[A]^2$

$\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

66) $\frac{d[O_2]}{dt} = \frac{0.032}{32}$ mol/hr

$= 10^{-3}$ mol/hr

$\frac{-d[N_2O_5]}{dt} = \frac{2d[O_2]}{dt}$

$\frac{-d[N_2O_5]}{dt} = 2 \times 10^{-3}$ mol/hr

$= 2 \times 10^{-3} \times 108$ g/hr

$= 0.216$ g/hr

67)

$r = k[NO_2]^1$

$t_{1/2} \propto a^0$

68) $k = \frac{1}{t}\ell n\left(\frac{v_\infty - v_0}{v_\infty - v_t}\right)$

$k = \frac{1}{20}\ell n\left(\frac{70-0}{70-40}\right)$

$k = \frac{1}{20}\ell n\frac{70}{30}$

$k = \frac{1}{20}\ell n\frac{7}{3}\ min^{-1}$

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69)

For a reaction, the rate constant k increases with temperature according to the Arrhenius equation: $k = Ae^{-\frac{E_a}{RT}}$ where: - k is the rate constant, - A is the pre-exponential factor, - E_a is the activation energy, - R is the gas constant, - T is the absolute temperature (in Kelvin).

Half-life for zero-order reaction: $t_{1/2} = \frac{[A]_0}{2k}$ For first-order reactions: $t_{1/2} = \frac{0.693}{k}$ where $[A]_0$ is the initial concentration.

$T\uparrow$, $K\uparrow$, $t_{1/2}\downarrow$

70)

$$A \rightarrow B + C$$

$t = 0 \quad P_i$

$t = t \quad P_i - x \quad x \quad x$

$$P_i - x + 2x = P_t$$

$$x = P_t - P_i$$

$$k = \frac{1}{t}\ell n\frac{P_i}{P_i - (P_t - P_i)}$$

$$k = \frac{1}{t}\ell n\frac{P_i}{2P_i - P_t}$$

$$k = \frac{2.303}{t}\log\frac{P_i}{2P_i - P_t}$$

71) $$t = \frac{2.303}{k}\log\frac{(R)_0}{[R]}$$

$$\log\frac{[R_0]}{[R]} = \frac{kt}{2.303}$$

$$\text{Slope} = \frac{k}{2.303}$$

72)

$r = k[A]^x [B]^y$

$$\frac{6 \times 10^{-3}}{2.4 \times 10^{-2}} = \left(\frac{0.1}{0.4}\right)^x$$

$$\left(\frac{1}{4}\right)^x = \frac{1}{4} \Rightarrow x = 1$$

Now, $$\frac{7.2 \times 10^{-2}}{2.88 \times 10^{-1}} = \left(\frac{2}{4}\right)^y$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^y$$

$y = 2$

$r = k[A]^1 [B]^2$

73) $$r = -\frac{1}{2}\left[\frac{0.4 - 0.5}{10 - 0}\right]$$

$$r = +\frac{1}{2} \times \frac{0.1}{10} = 0.5 \times 10^{-2} \text{ M min}^{-1}$$

74)

$r = k[A]^0$, for zero order reaction

$\boxed{r = k}$

$t_{3/4} = \frac{3}{2} t_{1/2}$

$[A]_t = [A]_0 - Kt$

Slope = – K

$t_{1/2} = [A]_0/2K$

T↑ K↑ $t_{1/2}$↓

75) $t_{99.9\,\%} = 3t_{90\,\%}$

76)

y = 45 = ΔH

77)

	2X	→	Y
t = 0	a		
t = t	a – 2x		x

a – 2x = x

a = 3x

x = a/3

78) $r \propto \frac{I}{E_a}$

79) total time (t) $= (t_{1/2})_A \times 3 = 0.2 \times 3 = 0.6$ sec

For A : No. of half lives (n) = 3

For B : No. of half lives (n) $= \frac{t}{(t_{1/2})_B} = \frac{0.6}{0.3}$

⇒ n = 2

$\frac{[A]_t = \frac{[A]_0}{2^n}}{[B]_t = \frac{[B]_0}{2^n}} \Rightarrow \frac{[A]_t}{[B]_t} = \frac{3/8}{2/4} = \frac{3}{4}$

80)

For zero order reaction r = k

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81) $\frac{-d[A]}{2dt} = \frac{-d[B]}{dt} \Rightarrow \frac{-d[B]}{dt} = \frac{1}{2} \times 2 \times 10^{-4}$
$= 10^{-4}$

82) For endothermic reaction
$\Delta H = E_f - E_b > 0$
$E_f > E_b$

83)

Zero order reaction must be a complex reaction.

84)

For zero order reaction
$r = k[A]^0$
$r = k$

85) $\because t_{1/2} \propto \frac{1}{[A]_0}$
So, second order reaction

86) $r \propto \frac{1}{E_a}$

87)

For elementary reaction, order is same as molecularity.
Molecularity is always positive.

88) $e^{-E_a/RT} = x$
$\frac{-E_a}{RT} = 2.303 \log_{10} x$
$\log_{10} x = \frac{-Ea}{2.303RT}$
$\log_{10} x = \frac{-209.5 \times 10^3}{2.303 \times 8.314 \times 581}$
$\log_{10} x = -18.83$
$x = 10^{-18.83}$
$x = 10^{0.17} \times 10^{-19}$
$x = 1.5 \times 10^{-19}$

89) $\log\left(\frac{k_1'}{k_1}\right) = \frac{E_1}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$

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$$\log\left(\frac{k_2'}{k_2}\right) = \frac{E_2}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

For $E_1 > E_2$

$$\frac{k_1'}{k_1} > \frac{k_2'}{k_2}$$

90)

$r' = K[4A]^{1/2}\,[2B]^2$

$r' = 2 \times 4 \times K[A]^{1/2}\,[B]^2$

$r' = 8r$

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